

Where Does a Physical Constraint Belong in a Generative Model?

A Codimension-Controlled Benchmark for Constraint-Exact Power-Grid Scenario Generation

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Abstract

Generative models are increasingly used to produce operational scenarios for power systems, and such scenarios must satisfy the physical laws they describe: a set of injections that violates Kirchhoff’s laws is not a conservative forecast but an impossible one. Several methods achieve *exact* satisfaction of affine physical invariants, and the literature disagrees about which to prefer. We argue the open question is not whether to enforce a constraint but **where it belongs**: in the hypothesis class (train-time projection), at inference (zero-shot correction), in the loss (penalty), or in the coordinates (nullspace or completion). We give the affine theory — train-time projection is exact under any Runge–Kutta scheme, excludes no minimiser, and never increases the flow-matching loss — and then build a benchmark that varies one quantity, the **codimension** of the constraint set, across 16 methods, 10 seeds and three suites derived from real grid measurements. The answer is not universal. On a controlled contrast in which the only change is how much of the state the physics determines, train-time projection moves from statistically indistinguishable from unconstrained flow matching (-0.07% , $t = -0.26$) to a significant improvement (-1.96% , $t = -6.29$); on a third real dataset the identical code is significantly worse ($+3.71\%$, $t = +5.50$). Two further results cut against common practice. Soft penalties are dominated on both axes: raising λ from 1 to 1000 moves the worst-case violation only from 219 MW to 187 MW while degrading the energy score by $+92\%$. And in a two-stage stochastic unit-commitment study, scenario feasibility does not reach the scheduling decision: cost regret correlates -0.933 with interval coverage but only $+0.315$ with the energy score ($p = 0.39$, not significant). All claims were pre-registered with explicit falsification conditions before the sweep; two of our own predictions were falsified by our own data and are reported as falsified.

Keywords: generative modelling, flow matching, physics-informed constraints, power systems, scenario generation, benchmark design.

1 Introduction

Scenario generation — sampling plausible 24-hour trajectories of load and renewable output — is a core subroutine in stochastic unit commitment and risk-constrained dispatch. Diffusion models [2, 3] and flow matching [1] produce high-fidelity samples, but a sample that violates power balance is not merely inaccurate: it is inadmissible as input to an optimiser that assumes it.

Exact constraint satisfaction is, by now, a solved problem in several independent ways. One can project the velocity field at training time, correct the sample at inference [5], complete the state from free coordinates [4], or work in a nullspace basis. Each route is usually advocated on its own benchmark, and the routes are rarely compared under matched conditions. **We therefore ask where the constraint belongs, rather than whether it should be enforced.**

Contributions.

- **Affine theory.** Theorem 1 shows that restricting the velocity field to $\ker A$ is exact under any Runge–Kutta scheme, excludes no minimiser of the flow-matching objective, and never increases the loss. Exactness costs no function evaluations.
- **A codimension-controlled benchmark.** Three suites built from real EIA-930 measurements and the pglib IEEE 118-bus network [16, 17], differing in the fraction of the state the physics determines, with two suites identical except for that fraction.
- **The empirical answer, which is not universal.** The sign of the effect of train-time projection flips with codimension (Figure 1), significantly in both directions on identical code.
- **A decision-level evaluation.** Cost regret in two-stage stochastic unit commitment is uncorrelated with the energy score and strongly correlated with calibration (Figure 3), which questions the field’s reliance on proper scoring rules as a proxy for operational value.

What we do not claim. We claim no novelty for hard constraints in flow matching: [5] imposes arbitrary nonlinear constraints zero-shot at inference. We claim no novelty for adaptive-step sampling [9, 8]. Our method is not symplectic and is unrelated to Hamiltonian generative flows [6] or Hamiltonian neural networks [7]; we report NFE and analytic FLOPs rather than device joules, because those are exactly countable and hardware-independent.

2 Where a Constraint Can Go

Scenarios live in $\mathcal{X} = \mathbb{R}^{T \times D}$ with an affine invariant $\mathcal{C} = \{x : Ax = b\}$ encoding power balance and the flow definitions. Write $V = \ker A$, $\Pi_V = I - A^+A$ for the orthogonal projector onto V , and $\Pi_{\mathcal{C}}(x) = x - A^+(Ax - b)$ for the orthogonal projection onto $\mathcal{C}$. Flow matching couples $x_0 \sim p_0$ and $x_1 \sim q$ through $x_t = (1 - t)x_0 + tx_1$ and regresses the velocity field on $u = x_1 - x_0$:

$$\mathcal{L}_{\text{FM}}(v) = \mathbb{E}_{t, x_0, x_1} \|v(x_t, t) - (x_1 - x_0)\|_2^2. \quad (1)$$

The four routes differ in where $\mathcal{C}$ is imposed:

$$\underbrace{\mathcal{L}_{\text{FM}}(v) + \lambda \|Ax - b\|^2}_{\text{penalty}}, \quad \underbrace{\mathcal{L}_{\text{FM}}(\Pi_V v)}_{\text{train-time projection}}, \quad \underbrace{\Pi_{\mathcal{C}} \circ \text{ODE}[v]}_{\text{post-hoc / inference-time}}, \quad \underbrace{x = x_p + Nc}_{\text{chart / completion}}. \quad (2)$$

Theorem 1 (Exact, free, and bias-free). *Let q be supported on $\mathcal{C}$ and $p_0 = \Pi_{\mathcal{C}\#} \mathcal{N}(0, I)$. Then (i) $x_t \in \mathcal{C}$ for all t and $u \in V$ almost surely; (ii) the flow-matching optimum $v^*(x, t) = \mathbb{E}[u \mid x_t = x]$ is V -valued, so restricting the hypothesis class to V -valued fields excludes no minimiser; (iii) for every v with finite second moment, $\mathcal{L}_{\text{FM}}(\Pi_V v) = \mathcal{L}_{\text{FM}}(v) - \mathbb{E}\|(I - \Pi_V)v\|^2 \leq \mathcal{L}_{\text{FM}}(v)$, with equality iff v is V -valued; and (iv) if $x \in \mathcal{C}$ and an integrator forms $x^+ = x + h \sum_i b_i k_i$ with every stage k_i an evaluation of a V -valued field, then $x^+ \in \mathcal{C}$ — for any step size, stage count or order.*

Proof sketch. (i) $\mathcal{C}$ is convex and contains x_0, x_1 , and $Au = b - b = 0$. (ii) V is a closed subspace and $u \in V$ a.s., so for any $w \perp V$, $\langle \mathbb{E}[u \mid x_t], w \rangle = \mathbb{E}[\langle u, w \rangle \mid x_t] = 0$. (iii) Decompose $v = \Pi_V v + (I - \Pi_V)v$; since

Table 1: **Benchmark geometry.** D is channels per hour; r is the rank of A per hour.

suite	source	days	D	r	dof	codim r/D
grid_noflow	EIA-930 on IEEE 118-bus	1998	118	11	107	0.093
measured	EIA-930, 6 balancing authorities	1990	69	21	48	0.304
grid	as grid_noflow + 186 line flows	1998	304	197	107	0.648

$u \in V$ the cross term vanishes, giving the Pythagorean identity. (iv) $Ax^+ = Ax + h \sum_i b_i(Ak_i) = Ax = b$ because $Ak_i = 0$; induct over steps. Full proofs are in the supplement. $\square$

Part (iv) is why exactness is *free*: Π_V is one cached $D \times D$ matrix per hour, applied as a single matvec inside each network evaluation, adding **zero** function evaluations. Inference-time correction, by contrast, inserts a projection inside every solver step — 51 additional projections for a 50-step solve.

Proposition 1 (Drift on a nonlinear manifold). *Let $\mathcal{M} = \{x : g(x) = 0\}$ with $g \in C^2$ and $J = \nabla g$ of full row rank. For a tangential field ($Jv = 0$ on $\mathcal{M}$), an order- p Runge–Kutta step leaves $\|g\|$ unchanged to $O(h^{p+1})$, so the invariant drifts by $O(h^p)$ over $1/h$ steps; one Gauss–Newton retraction $R(x) = x - J^+g(x)$ contracts quadratically [10]. Section 5.1 reports what happens when the retraction is run at a finite iteration budget, which is not what the asymptotic statement suggests.*

3 Benchmark

All suites are built from public data with no API key: EIA-930 hourly balancing-authority operating data (2019-01-02 to 2024-06-30) and the pglib-opf IEEE 118-bus and MATPOWER case57 networks [16, 17, 20]. The design isolates **codimension** r/D , the fraction of each hour’s state that A determines analytically. Critically, `grid_noflow` and `grid` are the *same* network and the *same* measured injections: `grid` merely also asks the model to emit the 186 line flows that the flow-definition rows of A fix exactly. The contrast is therefore causal rather than correlational.

We compare 16 generators over 10 seeds with an identical backbone, training budget and data: unconstrained flow matching; penalty variants at $\lambda \in \{1, 10, 100, 1000\}$; post-hoc projection; inference-time correction in the style of [5]; DC3-style completion [4]; a reduced nullspace chart; train-time projection (ours); and classical and deep baselines — Gaussian copula, k -NN historical resampling, cVAE, cWGAN-GP [18], RealNVP-style normalising flows [14, 15] and DDPM [2]. Scoring uses strictly proper rules — energy score and variogram score [11, 12, 13] — plus calibration and worst-case constraint violation.

4 Results

4.1 The sign of the constraint effect flips with codimension

Table 2 gives the numbers. At codimension 0.093 train-time projection is indistinguishable from doing nothing (-0.07% , $t = -0.26$); at 0.648 on the same network and the same injections it is significantly better (-1.96% , $t = -6.29$). On `measured` it is significantly worse ($+3.71\%$, $t = +5.50$).

What the third suite does and does not establish. `measured` is a different data-generating process — EIA-930 accounting identities across six balancing authorities rather than Kirchhoff’s laws on a synthetic

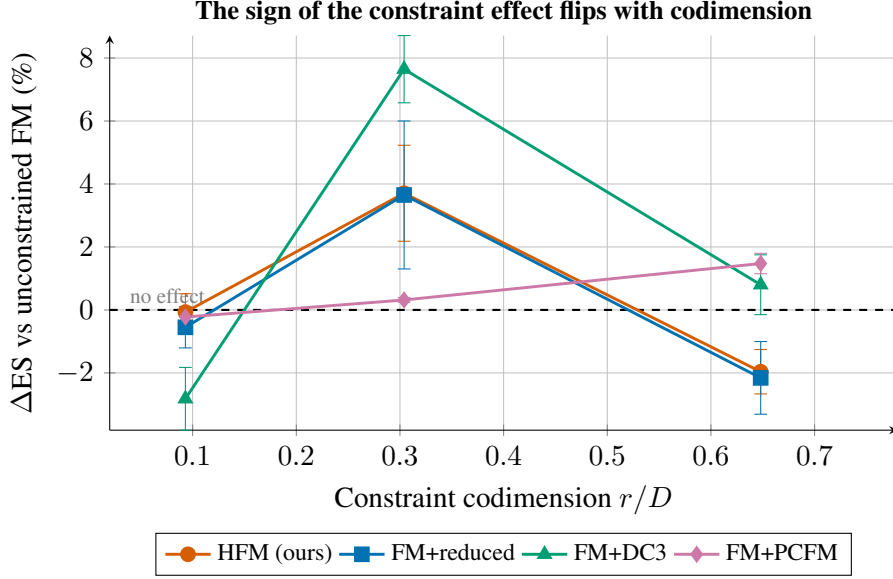


Figure 1: **The paper’s central result.** Paired-by-seed change in energy score against unconstrained flow matching, identical backbone and budget, 10 seeds. Points below zero are improvements. Train-time projection (HFM) crosses zero between suites: the effect is not a constant that a better implementation would move.

network — so it is a confounded third point, not a third rung of one ladder. We therefore do *not* claim that codimension alone predicts the magnitude of the effect. What it does establish is that real problems exist on which exact projection significantly *costs* fidelity, which is enough to refute any universal recommendation. We checked and rejected one alternative explanation: channel scale heterogeneity does not account for the pattern, since `gridnoflow` has the widest scale spread ($36\times$) and shows no effect.

4.2 Soft penalties are dominated on both axes

Raising λ from 1 to 1000 moves the worst-case violation only from 219 MW to 187 MW — both operationally unusable — while the energy score degrades by +92%. Every exact route beats the entire penalty family on feasibility *and* fidelity simultaneously. Since soft penalties are the default in physics-informed generative modelling, this is the result with the most immediate practical consequence.

4.3 Feasibility is not decision value

Across the 16 generators scored on both, cost regret correlates -0.933 with 90% interval coverage but only $+0.315$ with the energy score (Spearman $\rho = +0.229$, $p = 0.39$, not significant). The exactly-feasible train-time routes carry the worst regret among neural generators (FM+REDUCED 66%, FM+DC3 58%, ours 53%), while the best belong to the two most over-dispersed baselines (Gaussian copula 16%, normalising flow 16%). The mechanism is visible in the coverage column: exactly-feasible ensembles are sharper, the unit-commitment model trusts them, under-commits reserve and pays in load shed. **A study that stopped at the proper scoring rule would have reported the opposite conclusion with confidence.** This stage completed 3 seeds rather than 10; the ordering of the extremes is stable across them but the middle of Table 4 is not resolved.

Table 2: **Paired comparisons against unconstrained flow matching.** Positive ΔES means the constrained route scores *worse*. $|t| > 2.262$ is significant at the 5% level (9 d.o.f.).

suite	route	codim	ΔES (%)	paired t	verdict
grid_noflow	HFM (ours)	0.093	-0.07	-0.26	n.s.
grid_noflow	FM+reduced	0.093	-0.55	-1.91	n.s.
grid_noflow	FM+DC3	0.093	-2.82	-6.40	better
grid_noflow	FM+PCFM	0.093	-0.22	-2.70	better
grid_noflow	FM+posthoc	0.093	-0.00	-0.06	n.s.
measured	HFM (ours)	0.304	+3.71	+5.50	worse
measured	FM+reduced	0.304	+3.65	+3.52	worse
measured	FM+DC3	0.304	+7.65	+16.20	worse
measured	FM+PCFM	0.304	+0.32	+9.34	worse
measured	FM+posthoc	0.304	+0.26	+6.87	worse
grid	HFM (ours)	0.648	-1.96	-6.29	better
grid	FM+reduced	0.648	-2.16	-4.23	better
grid	FM+DC3	0.648	+0.80	+1.90	n.s.
grid	FM+PCFM	0.648	+1.47	+10.28	worse
grid	FM+posthoc	0.648	+1.06	+8.37	worse

Table 3: **Penalty sweep on grid.** Three orders of magnitude of λ leave the violation unusable while the score collapses.

λ	ES	rel. to FM (%)	$\ Ax - b\ _\infty$ (MW)
1	545.5	+0.3	219
10	595.5	+9.5	236
100	843.9	+55.1	277
1000	1043	+91.8	187
train-time projection	533.3	-2.0	6.48×10^{-4}

4.4 Exactness is free in function evaluations

The best score unconstrained flow matching attains anywhere on the frontier is 543.7 at NFE 548 (dopri5). Train-time projection reaches 539.5 at **NFE 20**, a $27\times$ reduction for a better score, holding the violation at 5.34×10^{-4} MW against 457 MW. Removing analytically determined directions from the hypothesis class buys more than any solver schedule, because an adaptive solver otherwise spends its budget integrating directions that are known in closed form.

5 Pre-Registered Predictions That Failed

Every claim was written into a pre-registration with an explicit falsification condition *before* the sweep ran, so acceptance criteria could not drift to fit the output. Two conditions fired, and we report them rather than softening them.

C3: a universal ordering of routes. We predicted $\text{train-time} \leq \text{inference-time} \leq \text{post-hoc}$ on fidelity, universally. The ordering holds on *grid* (-3.39% for train-time against inference-time, $t = -9.09$) and reverses on *measured* ($+3.38\%$, $t = +5.05$). The reversal became the central finding of this paper.

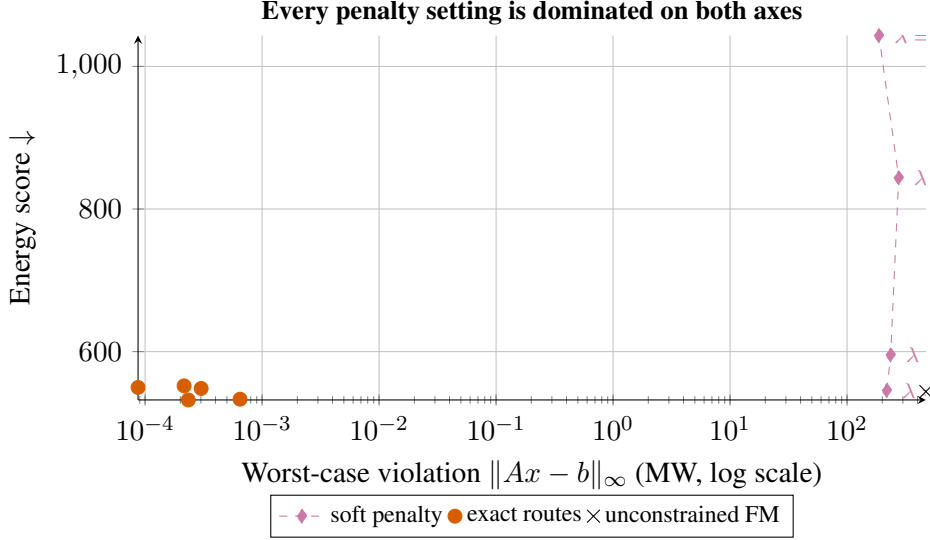


Figure 2: **There is no λ to tune.** Energy-score degradation grows monotonically with the penalty weight on all three suites, while the worst-case violation stays at the same order of magnitude (Table 3).

C5': physical versus chart coordinates under $N - 1$. We predicted that routes operating in physical coordinates would retain their learned distribution when an outage changes the PTDF, whereas chart-based routes would not. Measured over eight single-branch outages, ours attains ES 896.4 against 881.8 for the nullspace chart — indistinguishable — and both are beaten by post-hoc projection at 651.5. Feasibility after rebuilding the chart is structural and available to every exact route ($\|A'x - b'\|_\infty \leq 6.9e - 04$ MW). Per the pre-registration, the transfer differentiator is dropped.

5.1 A negative result on the nonlinear manifold

Proposition 1 suggests retracting every step. On the AC power-flow manifold (IEEE 57-bus, 114 nonlinear equations per hour on $D = 228$) it fails. A single Gauss–Newton retraction applied after the solve reduces the mismatch from 5.1 to 0.0077 at no cost in score. Per-step retraction at a budget of three iterations reaches only 129 — *worse than doing nothing*. Raising the budget to 20 iterations does not rescue it: the mismatch is $1.84e+03$, so the failure is not an under-resourced retraction but the in-loop pull-back injecting a biased correction at every step. We report this rather than omitting it.

6 Discussion

What practitioners should take away. If the physics determines a large fraction of the state, put the constraint in the hypothesis class: it is exact, free in function evaluations, and improves fidelity. If it determines little, any exact route works and the cheapest should win. Do not use a soft penalty. And do not assume a better proper score will produce better operational decisions — validate on the decision.

A classical baseline is competitive. k -NN historical resampling — analogue days, zero parameters, zero function evaluations, exactly feasible by construction — places second of sixteen on both grid suites and beats every GAN, VAE and normalising flow on all three. We pre-committed to reporting this if it happened.

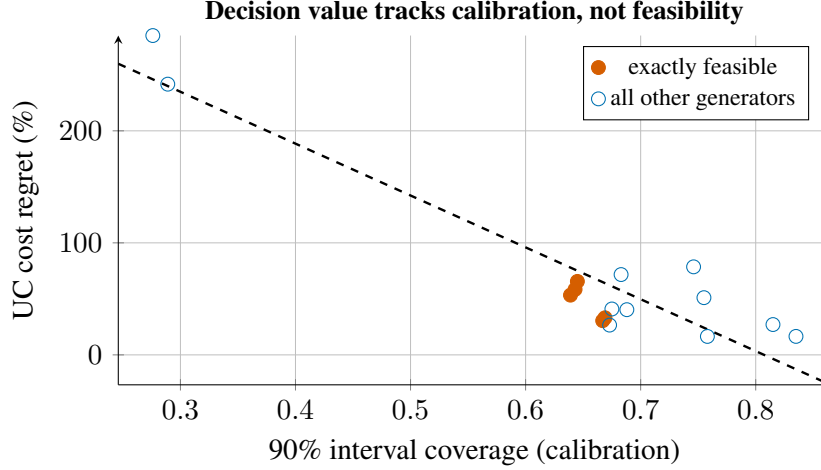


Figure 3: **Cost regret tracks calibration, not feasibility.** Two-stage stochastic unit commitment [19]; commitment frozen on the generated scenarios and scored on the realised day. Each marker is one generator. The exactly-feasible routes sit at the high-regret end.

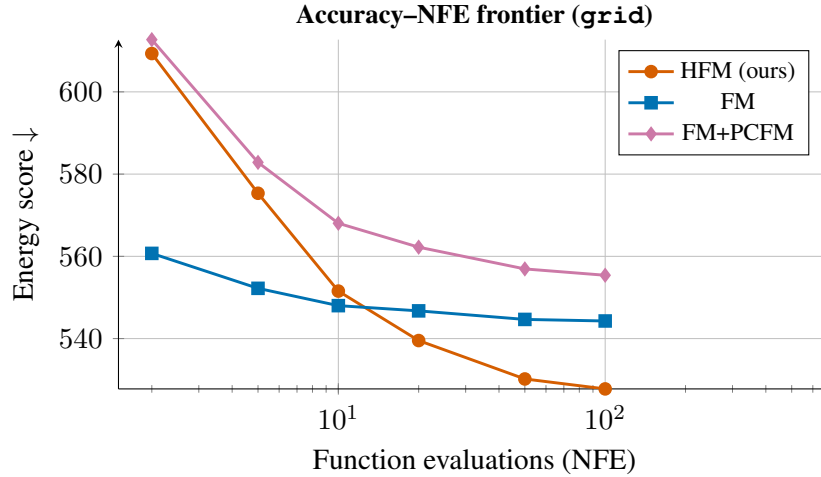


Figure 4: **Accuracy–NFE frontier.** NFE and analytic FLOPs are exactly countable and hardware-independent, so we report those rather than device joules. Train-time projection dominates at every budget.

Limitations. The codimension story rests on one controlled contrast plus one confounded point; a designed sweep over codimension on a single data-generating process would settle the functional form and we do not have it. The downstream stage has 3 seeds rather than 10. The AC results cover one network. Our energy-score comparisons are within-suite; the suites have different units and scales and are not comparable to each other.

7 Reproducibility

All data are public and require no API key. Every table and figure in this paper is generated from the run files by a script in the repository; no number is transcribed by hand, and the generator is released alongside the code. The pre-registration, including the falsification conditions that fired, is included in the supplement in the form in which it was written before the sweep.

Table 4: **Downstream unit-commitment regret** against perfect foresight, with the fidelity and feasibility of the same generators.

generator	regret (%)	$\pm$	cov ₉₀	$\ Ax - b\ _\infty$
NormFlow-RealNVP	16.5	4.3	0.758	1.81×10^5
GaussianCopula	16.5	3.2	0.835	4.33×10^4
FM	26.6	11.0	0.673	2.38×10^4
kNN-Historical	27.1	8.8	0.815	0
FM+PCFM	30.5	7.2	0.667	0.0348
FM+posthoc	33.1	10.1	0.669	0.028
FM+penalty(10)	40.3	4.9	0.688	1.38×10^4
FM+penalty(1)	41.0	13.3	0.675	1.91×10^4
DDPM	51.1	12.1	0.755	1.50×10^4
HFM (ours)	53.3	3.7	0.639	0.0622
FM+DC3	58.5	4.8	0.643	0.0375
FM+reduced	65.6	6.0	0.645	0.0555
FM+penalty(100)	71.7	9.5	0.683	2.23×10^4
FM+penalty(1000)	78.6	11.7	0.746	1.53×10^4
cVAE	241.7	91.7	0.289	7×10^3
cWGAN-GP	285.1	124.7	0.276	8.09×10^3

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Table 5: **AC manifold.** $|g|_\infty$ is the worst-case power-flow mismatch.

method	retract	iters	steps	$ g _\infty$	ES
FM	3	5	5	5.1	2.0371
FM	3	10	10	5.1	2.0018
FM	3	20	20	5.1	1.9850
FM	3	50	50	5.11	1.9757
FM	3	100	100	5.11	1.9730
FM+retract-final	3	5	5	0.00785	2.0429
FM+retract-final	3	10	10	0.00768	2.0044
FM+retract-final	3	20	20	0.00762	1.9862
FM+retract-final	3	50	50	0.00767	1.9762
FM+retract-final	3	100	100	0.00767	1.9732
Manifold-FM (ours)	3	5	5	129	1.9953
Manifold-FM (ours)	3	10	10	129	1.9514
Manifold-FM (ours)	3	20	20	129	1.9168
Manifold-FM (ours)	3	50	50	129	1.9040
Manifold-FM (ours)	3	100	100	129	1.8966
Manifold-FM (ours)	20	5	5	129	1.9952
Manifold-FM (ours)	20	10	10	3.56×10^3	1.9511

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